Software Requirements Specification

for

AI-Based Attendance Monitoring System

Prepared by:

Arjun Agrawal (2400290310042)

Pankaj Bhargav (2500290109018)

Pankaj Chaudhary (2500290109019)

Prachi (2500290109020)

Department of Computer Science & Engineering

KIET Group of Institutions

Academic Project / Software Engineering Lab

September 2026

Table of Contents

Revision History 2
1. Introduction 3
1.1 Purpose 3
1.2 Document Conventions 3
1.3 Intended Audience and Reading Suggestions 3
1.4 Product Scope 3
1.5 References 3
2. Overall Description 4
2.1 Product Perspective 4
2.2 Product Functions 4
2.3 User Classes and Characteristics 4
2.4 Operating Environment 4
2.5 Constraints, Assumptions and Dependencies 4
3. External Interface Requirements 5
3.1 User Interface 5
3.2 Hardware Interface 5
3.3 Software Interface 5
3.4 Communication Interface 5
4. System Features and Functional Requirements 6
4.1 Student Enrollment 6
4.2 Attendance Capture and Recognition 6
4.3 Review Queue and Correction 6
4.4 Timetable and Notifications 6
5. Other Nonfunctional Requirements 7
5.1 Performance 7
5.2 Security and Privacy 7
5.3 Reliability and Usability 7
6. Data Requirements and Business Rules 8
7. Acceptance Criteria and Future Enhancements 9
Appendix A: Glossary 9

Revision History

Version	Date	Author/Team	Reason for Changes
1.0	September 2026	Project Team	Initial SRS prepared for the AI-Based Attendance Monitoring System.

This SRS follows the structure of the supplied IEEE-style SRS laboratory template and is tailored to the project's feasibility report.

1. Introduction

1.1 Purpose

This Software Requirements Specification defines the requirements for the AI-Based Attendance Monitoring System. The system is intended to automate classroom attendance using a camera, face detection, face recognition and a teacher review mechanism.

The SRS describes what the system should do, the users who will interact with it, its interfaces, functional requirements, data requirements, performance expectations and security/privacy considerations.

1.2 Document Conventions

·REQ-F-xxx denotes a functional requirement.
·REQ-NF-xxx denotes a non-functional requirement.
·REQ-SEC-xxx denotes a security/privacy requirement.
·REQ-BR-xxx denotes a business rule.
·Priority values are High, Medium, or Low.
·The terms SHALL/MUST indicate mandatory requirements; SHOULD indicates a recommended behavior.
·TBD indicates information that must be finalized during detailed design or deployment.

1.3 Intended Audience and Reading Suggestions

The document is intended for the student development team, project manager, software developers, testers, teachers and academic evaluators. Developers can focus on Sections 3–6; testers can use the functional and acceptance requirements; teachers and evaluators can focus on Sections 1, 2, 4 and 7.

1.4 Product Scope

The system covers student enrollment, camera-based face detection and recognition, liveness verification, confidence-based attendance marking, teacher correction, timetable-triggered sessions, attendance dashboards and basic reports. The pilot is intended for a classroom-scale deployment.

It does not replace the institution's complete academic ERP, perform disciplinary decisions automatically, or claim perfect recognition in all environmental conditions.

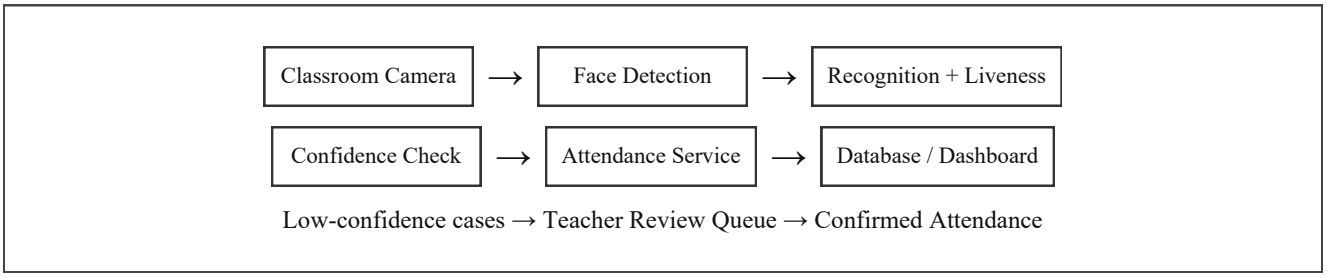
1.5 References

Experiment 1 — Software Requirements Specification, supplied IEEE-style laboratory template.
Experiment 2 — Feasibility Report for AI-Based Attendance Monitoring System.
Project team's proposed technical design and feasibility assumptions.

2. Overall Description

2.1 Product Perspective

The system is a new classroom attendance subsystem. A camera observes the classroom during an attendance session. Video frames are processed by the recognition pipeline, matched against enrolled student embeddings and sent to the attendance service. The teacher dashboard provides status and correction controls.



2.2 Product Functions

Enroll students using multiple reference images and generate facial embeddings.
Start attendance automatically according to the timetable or manually by a teacher.
Detect multiple faces across sampled frames and recognise enrolled students.
Run a lightweight liveness check before accepting a match.
Mark high-confidence matches and route uncertain cases to teacher review.
Maintain attendance history and provide class/student summaries.
Provide attendance streaks and low-attendance notifications.

2.3 User Classes and Characteristics

User	Responsibilities	Expected Skill
Teacher	Start/monitor sessions, review uncertain matches, correct attendance and view reports.	Basic web/app usage.
Student	Complete enrollment and view attendance/streak information.	Basic mobile/web usage.
Administrator	Manage classes, users, timetables, devices and system configuration.	Basic technical/admin skills.

2. Overall Description — Continued

2.4 Operating Environment

The pilot system is designed for a classroom with a fixed wide-angle camera, a local or network-connected processing machine and a browser-based teacher dashboard. A backend service may run on a classroom PC, server or suitable edge device.

Component	Proposed Environment
Client	Modern web browser on teacher/admin device.
Backend	Python service using Django REST Framework or FastAPI.
Computer Vision	OpenCV with face detection/recognition and liveness components.
Database	PostgreSQL or MySQL.
Camera	1080p minimum wide-angle camera; higher resolution preferred for distant rows.
Optional Edge Hardware	Raspberry Pi / Jetson-class device where local processing is required.

2.5 Design and Implementation Constraints

The camera must cover the intended seating area with acceptable face size and lighting.
Recognition accuracy depends on enrollment quality, camera position, lighting and occlusion.
The system must not treat a low-confidence prediction as a guaranteed identity.
Facial data requires controlled access, retention rules and privacy safeguards.
The pilot budget is constrained; the feasibility estimate considered approximately ₹13,000–₹30,000.

2.6 Assumptions and Dependencies

Students are enrolled before automated attendance is used.
The timetable is available and sufficiently accurate for automated session triggering.
The camera remains fixed after calibration.
Teachers can correct exceptions through the review queue.
Internet connectivity is available when cloud/database services are used; local processing can reduce this dependency.

3. External Interface Requirements

3.1 User Interfaces

The teacher dashboard shall show the active class, attendance status, recognised students, uncertain cases and controls for manual correction. The student view shall show attendance percentage/history and streak information. Administrative screens shall provide class, user, timetable and device management.

3.2 Hardware Interfaces

Camera shall provide a continuous or sampled video stream to the processing service.
Camera placement shall be calibrated to cover the classroom without unnecessary blind spots.
Optional edge hardware shall expose sufficient compute, memory and camera/network connectivity.

3.3 Software Interfaces

Interface	Purpose
Face Detection Module	Find multiple faces in sampled frames.
Recognition Module	Generate/compare face embeddings with enrolled identities.
Liveness Module	Reduce simple photo/video spoofing attempts.
Attendance API	Receive recognition events and create/update attendance records.
Database	Store users, classes, timetables, embeddings, attendance and audit records.

3.4 Communication Interfaces

The web dashboard shall communicate with the backend through HTTP/HTTPS APIs. The camera stream may use a local camera interface or network stream. Authentication shall be required for protected API operations.

3.5 Data Flow

Camera frames → sampled frames → face detections → embeddings → identity candidates → liveness/confidence check → attendance event → database → dashboard/report

4. System Features and Functional Requirements

4.1 Student Enrollment

FR-01 The system shall allow an authorised administrator/teacher to create a student profile.

FR-02 The system shall capture 3–5 reference images under reasonable lighting and angle variation.

FR-03 The system shall generate a facial embedding for the enrolled student.

FR-04 The system shall allow authorised users to re-enroll a student when recognition confidence repeatedly degrades.

4.2 Attendance Capture and Recognition

FR-05 The system shall create an attendance session from the timetable or teacher action.

FR-06 The system shall sample multiple frames during the initial attendance window rather than relying on a single snapshot.

FR-07 The system shall detect multiple faces in a frame and attempt recognition against enrolled students.

FR-08 The system shall perform a liveness check before accepting a recognised identity where the configured mode requires it.

FR-09 The system shall mark a student present only when the configured confidence threshold is satisfied.

FR-10 The system shall avoid duplicate attendance entries for the same student in one session.

4.3 Review Queue and Correction

FR-11 The system shall place low-confidence or unresolved cases in a teacher review queue.

FR-12 The teacher shall be able to confirm, reject or manually correct a review item.

FR-13 A teacher-confirmed low-confidence match may be used as a trusted signal for future embedding refresh, subject to system policy.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

NFR-01 The system should process sampled classroom frames quickly enough to complete the attendance workflow without creating significant class disruption.
NFR-02 The system should support multiple faces per frame and tolerate temporary occlusion by using multiple frames.
NFR-03 Dashboard actions such as opening a review queue or saving a correction should normally complete within a few seconds under normal pilot load.

5.2 Security and Privacy Requirements

NFR-04 Only authorised users shall access attendance records and administrative functions.
NFR-05 Facial embeddings and attendance records shall be protected against unauthorised access.
NFR-06 The system should minimise retention of raw face images and prefer embeddings for routine recognition storage.
NFR-07 The system shall maintain an audit trail for manual attendance corrections.
NFR-08 Access permissions shall follow the user's role.

5.3 Reliability and Usability

The system should continue sampling when an individual face is temporarily blocked or turned away.
If recognition is uncertain, the system should fail safely by requesting teacher review rather than making an unverified identity claim.
The dashboard should make absent, present and review states visually distinguishable and easy to correct.
Important operations should provide clear success/error feedback.

5.4 Software Quality Attributes

Attribute	Requirement
Accuracy	Use calibrated thresholds and multi-frame evidence; do not assume 100% recognition.
Maintainability	Separate camera, CV, API, database and UI modules.
Scalability	Architecture should allow additional classrooms/cameras later.
Usability	Teacher review and correction should take only a few simple actions.

6. Data Requirements and Business Rules

6.1 Core Data Entities

Entity	Important Fields
Student	Student ID, name, class/section, status, enrollment metadata.
Face Profile	Student ID, embedding reference, enrollment version, confidence metadata.
Class	Class ID, course, section, teacher.
Timetable	Class, day/time, room, attendance-session rule.
Attendance	Student, session, timestamp, status, source, confidence/review state.
Audit Log	User, action, timestamp, affected attendance record.

6.2 Business Rules

One student should have at most one final attendance status per class session.
High-confidence, live matches may be marked present automatically.
Low-confidence matches shall remain pending until teacher review.
A manual correction shall record the responsible user and time.
Confirmed attendance can be used for future model/profile adaptation only under configured policy.

7. Acceptance Criteria and Future Enhancements

The pilot will be considered acceptable when an authorised user can enroll students, start a scheduled/manual session, detect and recognise multiple enrolled students, prevent obvious photo spoofing through the configured liveness step, review uncertain cases, correct attendance and view saved records.

Future enhancements may include stronger liveness models, better handling of masks/occlusion, mobile access, institution-wide integration, analytics, multilingual support and more advanced edge deployment.

Appendix A: Glossary

Term	Meaning
Embedding	Numerical representation used to compare a face with enrolled profiles.
Liveness	Check intended to distinguish a live person from a simple presentation attack.
Confidence	Score/measure indicating how strongly a recognition result matches an enrolled identity.
Review Queue	List of uncertain recognition cases requiring teacher confirmation.